

Know what your API change can affect before you merge.

Cross-repository impact analysis for engineering teams.

01	02	03	04
Backend PR	API Contract	Consumer Repositories	Impact Evidence

Your CI knows whether the provider works. Does it know whether consumers still work?

1 Backend Repository

2 API Change

3 Provider CI ✓

4 PR Merged

5 Consumer Repository ✗

6 Production Incident

PROVIDER CONTRACT DIFF · CONSUMER CODE, UNCHANGED

```
- user.firstName  
+ user.first_name  
user.firstName
```

Provider tests can pass while downstream repositories still depend on the old API contract.

See downstream consumer impact during code review.

From API diff to concrete consumer evidence.

Contract Diff

Changed fields,
schemas, endpoints,
headers

Consumer Mapping

Downstream
repositories and relevant
consumers

Code Analysis

Inspect actual consumer
code

Evidence

Exact file, line, field
usage, risk

BACKEND CHANGE

```
- firstName  
+ first_name
```

CONSUMER · src/users.ts:42

```
user.firstName
```

LIKELY BREAK

HIGH RISK

Every finding shows the exact file and line, so you can verify it yourself.

Built to be checked, not taken on faith.

Not every API change is a breaking change.

Findings point to the exact consumer code that is affected, and skip changes that do not matter.

FALSE ALARM

Removed, but never read

Backend removes legacyBadge; consumer calls the endpoint but never reads the field.

NOT AFFECTED

GUARDED USAGE

A fallback handles removal

```
const badge = user.legacyBadge ??  
user.currentBadge ?? "DEFAULT";
```

FALLBACK PRESENT · LOW RISK

Find real breakage. Suppress irrelevant noise.

13-case pilot on our test set

13

Public PRs in our test set

3

Open-source ecosystems

11

Breaking changes caught

GITHUB · PULL REQUEST

Spring Boot / Java

Spring PetClinic

GITHUB · PULL REQUEST

Strapi / Node.js

GITHUB · PULL REQUEST

Fastify / TypeScript

13 public PRs on our Spring PetClinic, Strapi and Fastify forks under github.com/impact-gate. On this test set: 11 breaking changes caught, 0 false alarms, ~6.8s average end-to-end analysis. These are pilot results, not a universal accuracy claim or an SLA.

EXAMPLE

See it on a pull request.

1

A backend PR changes a contract

2

Impact Gate returns consumer evidence

File, line and risk.

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Built Impact Gate after hitting these challenges in distributed engineering systems.

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